Interplay of galaxy formation and the evolution of dark matter haloes in the cosmic web

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IUCAA

List of Publications

1. Velmani & Paranjape, 2023, “The quasi-adiabatic relaxation of haloes in the IllustrisTNG and EAGLE cosmological simulations”, **MNRAS**, 520(2):2867-2886. doi:10.1093/mnras/stad297

2. Velmani & Paranjape, 2024, “A self-similar model of galaxy formation and dark halo relaxation”, **JCAP**, doi:10.1088/1475-7516/2024/05/080,

3. Velmani & Paranjape, 2025, “Dynamics of the response of dark matter halo to galaxy evolution in IllustrisTNG”, **JCAP**, doi:10.1088/1475-7516/2025/02/006,

4. Velmani & Paranjape, 2024, “Role of astrophysical modeling on dark matter halo relaxation response at redshifts $z = 0$ and $z = 1$”, arXiv:2408.04864, **JCAP** *subm.*

Not included in this thesis

1. Ramakrishnan & Velmani, 2022, “Properties beyond mass for unresolved haloes across redshift and cosmology using correlations with local halo environment.”, *MNRAS*, doi:10.1093/mnras/stac2605

Overview of the talk

1. Background and motivation
2. Organisation of the thesis
3. Simulations and analysis techniques
4. Findings
5. Summary and future plans

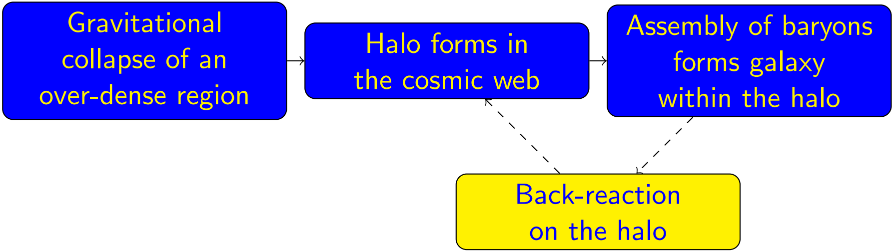
Introduction

- As matter clumps over overdensities due to gravity, the baryons cool and condense to form galaxies, leaving extended objects of dark matter haloes in the common paradigm of cosmology.
- These haloes are crucial in studying cosmology and dark matter particle physics, and are often modelled without any baryonic astrophysics using techniques like cosmological N-body simulations.
- Modeling galaxy formation does depend upon these haloes using for example semi-analytical techniques.

Gravitational collapse of an over-dense region → Halo forms in the cosmic web → Assembly of baryons forms galaxy within the halo

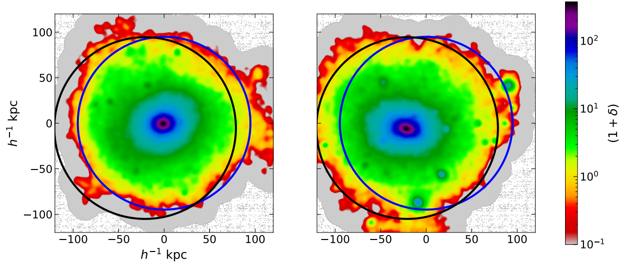
Introduction: Hydrodynamical simulations with galaxies

- Realistic galaxies are produced by modern hydrodynamical simulations of cosmological volumes with various subgrid baryonic prescriptions for the unresolved baryonic astrophysics such as in EAGLE, IllustrisTNG, Horizon, SIMBA, etc.
- In these simulations, the phase-space distribution of dark matter within the haloes have also been found to be significantly different and diverse indicating strong response to galaxies they host.



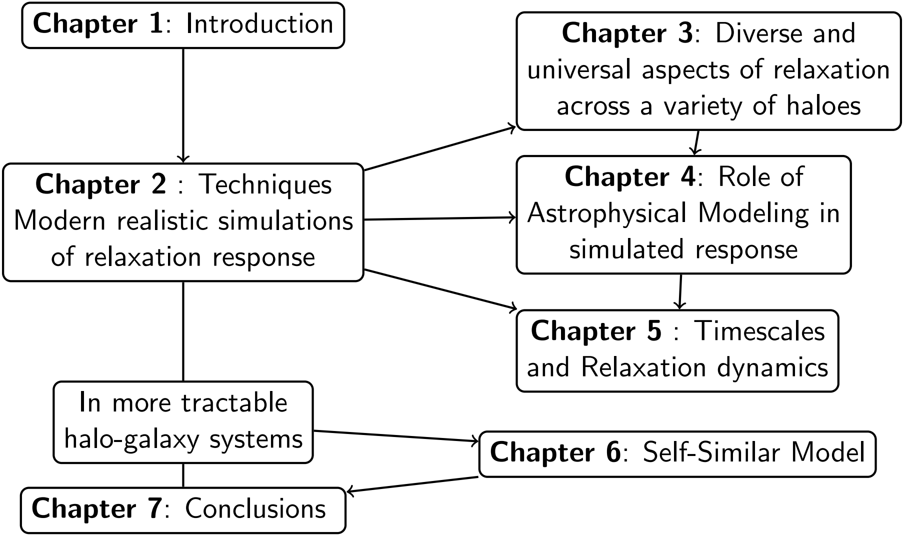
Dark matter halo response to galaxies

A halo from EAGLE simulation in the presence of galaxy (left image) can be seen more compact, spherical and even their centres shifted.



This thesis aims to understand the change in the radial mass distribution of dark matter in the haloes that influence key observables such as the rotation curves and radial acceleration relations.

Overview of thesis structure



Simulations

- IllustrisTNG and EAGLE

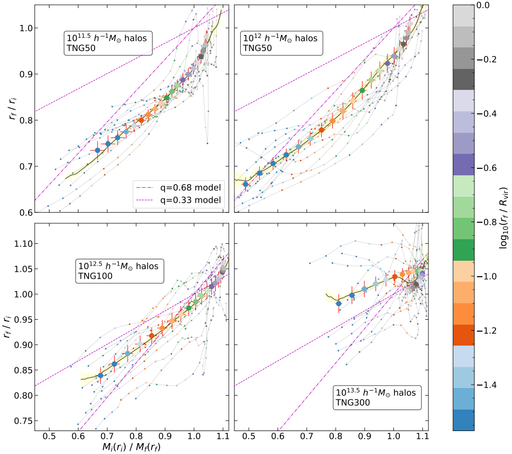
Physically characterizing the relaxation response

- Early works modeled this as adiabatic relaxation of dark matter in response to the **net** change in the gravitational potential due to galaxy formation.
- Consider a dark matter particle in circular orbit starting from radius r_i and adiabatically (conserving angular momentum) evolving to radius r_f caused by change in sphericalised total mass profile from $M_i(r)$ to $M_f(r)$ due to galaxy, then

$$r_i\, M_i(r_i) = r_f\, M_f(r_f) \implies \frac{r_f}{r_i} = \frac{M_i(r_i)}{M_f(r_f)}\,.$$

- By assuming no shell crossing, these quantities can be calculated by the comparing total and dark matter mass profile in the presence and absence of galaxies using the relation $M_f^d(r_f) = M_i^d(r_i)$.
- Quasi-adiabatic relaxation models consider the relaxation ratio r_f/r_i as a function of the mass ratio M_i/M_f.

Relaxation relation in IllustrisTNG

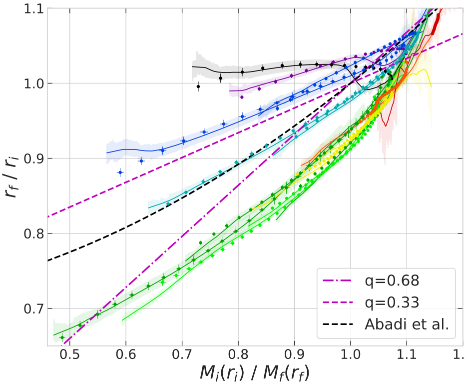


- Stacking the relaxation relation across haloes in populations selected by their sizes.
- Relaxation significantly weaker in cluster scale haloes.

Diversity in relaxation relation across halo masses

- Stacked relaxation relation is significantly different across halo masses.
- Existing quasi-adiabatic framework failed to provide a simple description across all scales.
- Relaxation significantly weaker in cluster scale haloes.

IllustrisTNG



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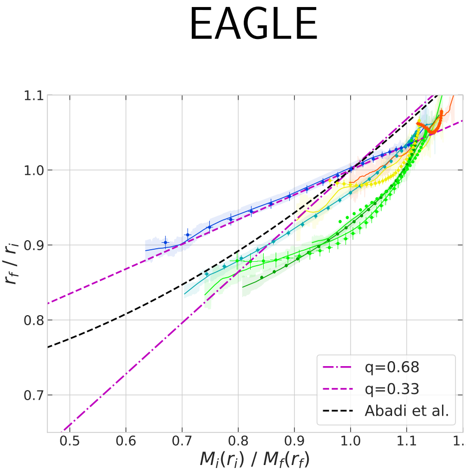
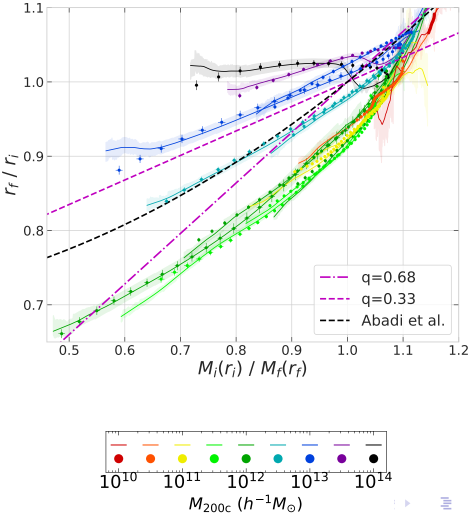


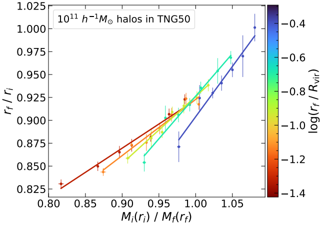
Relaxation in IllustrisTNG and EAGLE

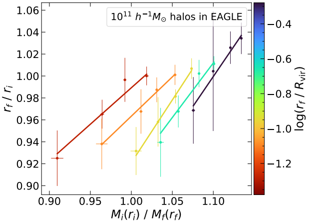
- We found that the relaxation relation (between r_f/r_i and M_i/M_f) varies widely between haloes of different mass scales in IllustrisTNG and EAGLE.
- Existing quasi-adiabatic framework failed to provide a simple description across all scales.

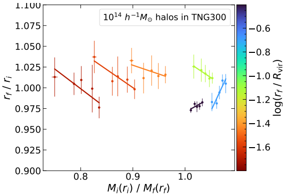


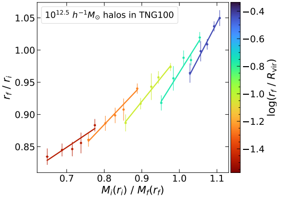
Dependence on halo centric distance

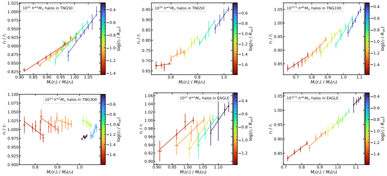
- A simple linear relation can accurately describe the relaxation, provided we assume an additional explicit dependence on the halo-centric distance.
- Notice that the exact nature of the relation is very different at different halo-centric distances (indicated by colorbar) for some halo masses, but they are all strongly linear.



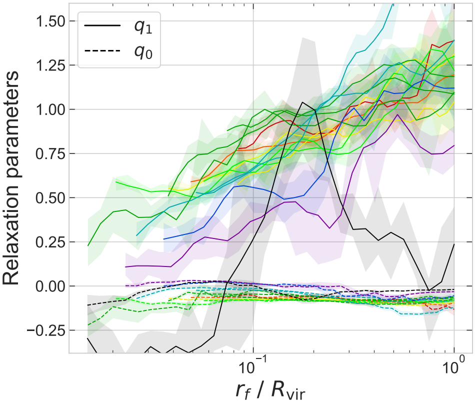




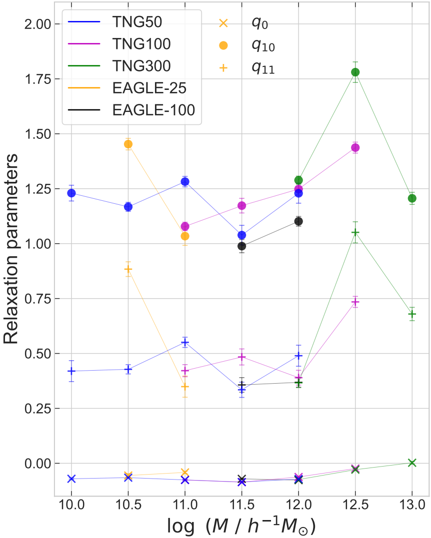




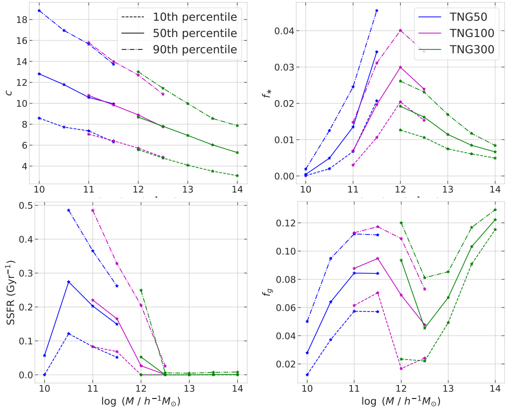
$10^{11}\ h^{-1}M_{\odot}$ halos in TNG50
$10^{12}\ h^{-1}M_{\odot}$ halos in TNG50
$10^{12.5}\ h^{-1}M_{\odot}$ halos in TNG100
$10^{14}\ h^{-1}M_{\odot}$ halos in TNG300
$10^{11}\ h^{-1}M_{\odot}$ halos in EAGLE
$10^{12.5}\ h^{-1}M_{\odot}$ halos in EAGLE
$r_f\ /\ r_i$
$M_i(r_i)\ /\ M_f(r_f)$
$\log(r_f\ /\ R_{vir})$



q_1
q_0
Relaxation parameters
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
10^{-1}
10^{0}
r_f / $R_{\rm vir}$

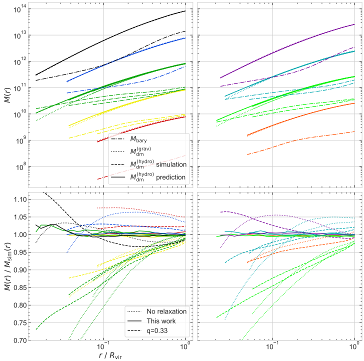

TNG50
TNG100
TNG300
EAGLE-25
EAGLE-100
q_0
q_{10}
q_{11}
Relaxation parameters
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
10.0
10.5
11.0
11.5
12.0
12.5
13.0
$\log\ (M\ /\ h^{-1}M_{\odot})$

- Percentiles



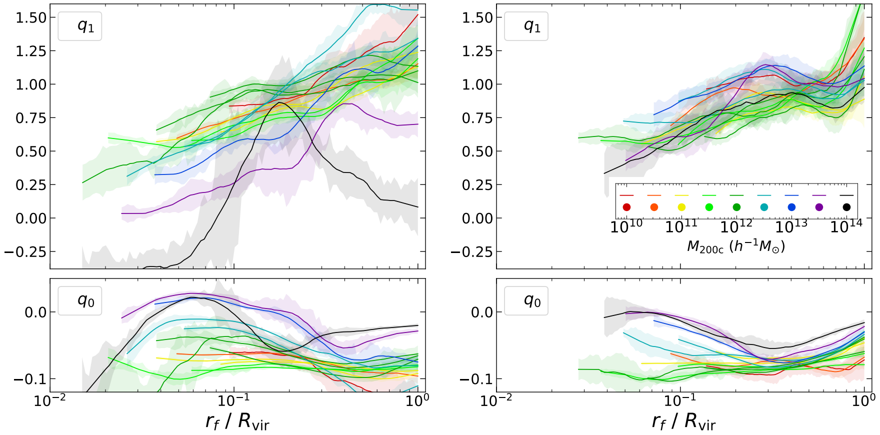
Applications to

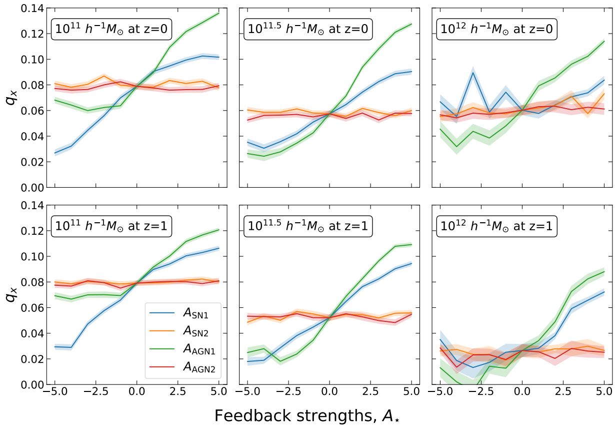
- Using the three parameter model, we can predict the relaxed dark matter profile given the baryonic mass profiles.
- These predictions are an order of magnitude better than existing models.



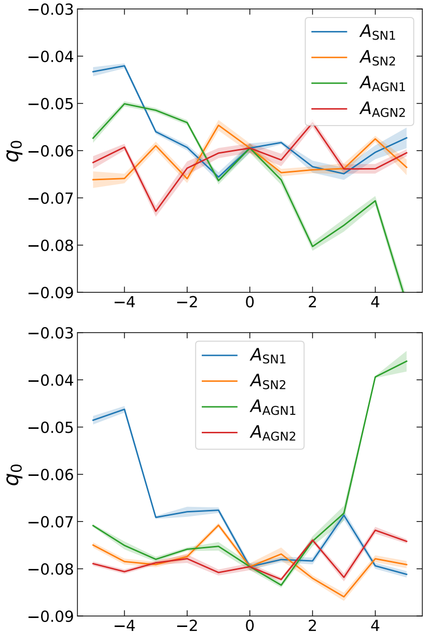
Universal description of Halo Relaxation Response at $z = 1$

The radially dependent slope $q_1(r_f)$ and intercept $q_0(r_f)$ of this linear fit, is more universal across a wide range of halo masses up to $10^{13}h^{-1}M_\odot$ at $z = 0$ (left) and up to $10^{14}h^{-1}M_\odot$ at earlier redshift, $z = 1$ (right).



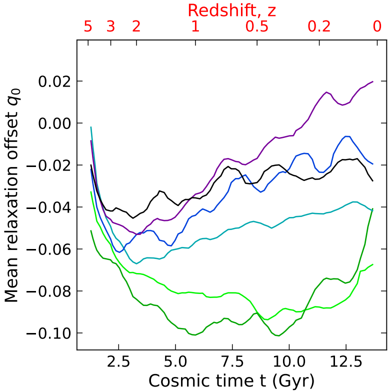

$10^{11}\ h^{-1}M_\odot$ at z=0
$10^{11.5}\ h^{-1}M_\odot$ at z=0
$10^{12}\ h^{-1}M_\odot$ at z=0
$10^{11}\ h^{-1}M_\odot$ at z=1
$10^{11.5}\ h^{-1}M_\odot$ at z=1
$10^{12}\ h^{-1}M_\odot$ at z=1
q_x
A_{SN1}
A_{SN2}
A_{AGN1}
A_{AGN2}
Feedback strengths, $A_\star$

- Relaxation offset parameter q_0 as a function of the baryonic feedback parameters in CAMELS-TNG. Left: $z = 1$, Right: $z = 0$.
- Using the suite of CAMELS simulations with variation in feedback strengths, we found that the offset parameter q_x, which is inversely related to q_0, indeed strongly depends on the overall feedback flux from the galaxies. However, the parameters controlling the feedback burstiness and the wind speed have negligible impact on the relaxation offset as shown below.



Relaxation Dynamics

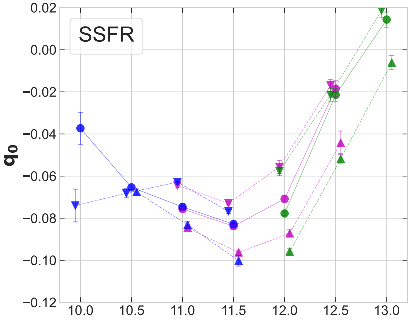
- Focusing on the intercept offset q_0, which describes the amount of relaxation $r_f/r_i - 1$ for dark matter shells with no net change in the total enclosed mass $M_i/M_f = 1$.
- It starts at zero, becomes more negative initially, but then apparently revert slowly back to zero.



Connection with star formation rate?

Astrophysical connection

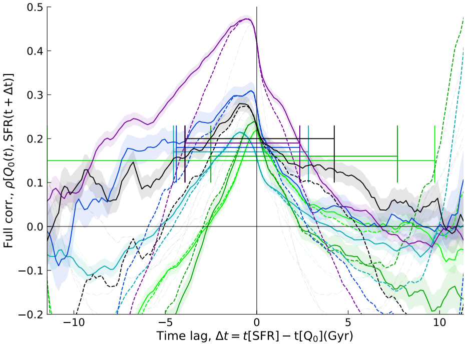
- Focusing on the intercept offset q_0, which describes the amount of relaxation $r_f/r_i - 1$ for dark matter shells with no net change in the total enclosed mass $M_i/M_f = 1$.
- The value of q_0 today was found to be more negative among haloes hosting galaxies with higher specific star formation rate (SSFR).



This higher excess relaxation might be related to larger amount of recent feedback output.

Temporal Connection with Astrophysics

Spearman-rank correlation between $Q_0 \equiv -q_0$ and mean SFR across a variety of haloes over long periods of time at different final halo masses suggests that the offset is typically stronger following periods of high star formation activity.



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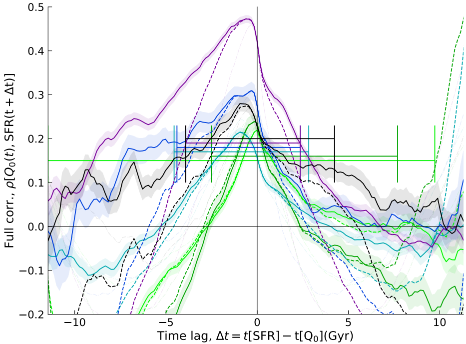

Additionally, we also found that the offset in the inner region is immediately affected by high SFR, while the outer regions are affected after around 2-3 Gyr. This likely explains the explicit dependence on the halo-centric distance.

Future plans

- Use direct probe of feedback to understand the role of AGN and other feedback on the relaxation dynamics.
- Use analytical tools and semi-analytical experiments to build an entirely physical and accurate model of relaxation.
- Run galaxy forming cosmological simulations specially focussed on studying the dark matter halo relaxation response.
- Identify galaxy properties that keeps record of evolutionary history of galactic processes allowing accurate determination of dark halo response.

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Thank You

For letting me share some of the correlations I made with the Universe, hope this increased your overall correlation with the elegant workings of the cosmos!

Introduction

Overview

Motivation

Realistic Simulations

Overview